Stage 3 — Questions and Answers

Feature Engineering | IPSS-R, Gene Panel and Centre-Grouped Validation

Q1. Score one patient with IPSS-R by hand

The karyotype **45,xy,-7[19]/46,xy[1]** contains monosomy 7 (loss of chromosome 7), which is one abnormality. The bracketed numbers are cell counts, not additional abnormalities. Monosomy 7 is in the **poor** cytogenetic category (3 points).

Component	Patient value	Points
Cytogenetics	Monosomy 7 — poor	3
Bone-marrow blasts	12% (>10%)	3
Haemoglobin	7.5 g/dL (<8)	1.5
Platelets	80 G/L (50 to <100)	0.5
ANC	0.6 G/L (<0.8)	0.5

> **Total = 3 + 3 + 1.5 + 0.5 + 0.5 = 8.5 points.**

A score greater than 6 is **very high risk**. In the training cohort, this group has a Kaplan–Meier median overall survival of **0.86 years** (about 10.3 months). This is a group median, not an individual survival prediction.

Q2. Why only use genes found in both centres?

Training contains 124 genes and test contains 83, with only 64 shared. EP300 mutations appear in 43 training patients but in zero test patients. That does not establish that no KYW patient carries an EP300 mutation: KYW may not have sequenced that gene. A model trained to associate EP300 with higher risk cannot reliably use its absence in KYW as evidence of lower risk.

The 598 training mutations outside the shared gene panel would also increase mutation counts for training patients relative to otherwise similar test patients if counted. Then **mol_n_mut** would partly measure sequencing-panel coverage rather than patient biology. Restricting mutation features to the 64 shared genes makes counts more comparable. As with the Stage 1 MONOCYTES missingness shift, a feature measured differently across centres can mislead a model.

Q3. What does ±0.030 tell us?

Across 15 centre-grouped validation folds, the fixed IPSS-R score has mean IPCW C-index **0.6709**, standard deviation **0.0299**, minimum **0.6240**, and maximum **0.7115**.

> **Range = 0.7115 − 0.6240 = 0.0875.**

If a Cox model scores 0.685, its improvement over the IPSS-R mean is:

> **0.685 − 0.671 = 0.014.**

Because 0.014 is smaller than the fold-to-fold spread of about 0.030, the higher mean is not yet convincing evidence of improvement. The fixed IPSS-R score changes across folds because the patient populations, clinical centres, observed deaths, and censoring differ. A stronger test is to compare Cox and IPSS-R **on the same folds** and examine the paired differences: does Cox beat IPSS-R consistently across most folds, with a stable improvement?

Sources: Stage 3 paper, sections 6.2–6.6; uploaded km_ipssr_groups.csv, ipssr_by_fold.csv and gene_panel.csv; provided 05_build_features.py output.